

ELITE MATHEMATICS

Pearson Edexcel International GCSE

Linear 4MA1

Adaptive Classified

OL-T09 · Rounding, Estimation and Bounds

Questions

Student book · questions with writing space

■ 23 pages

FAMILY MAP

Rounding Estimation Bounds

F01	Recover single-measurement bounds	7 Q 12 marks
F02	Optimise a compound expression using bounds	11 Q 35 marks
F03	Round to a stated accuracy	2 Q 10 marks
F04	Estimate a numerical calculation	2 Q 6 marks

Working Unit 01

4MA1-1H Q2

UNIT 1
2 marks

The height of each cup is 9 cm, correct to the nearest cm

(b) Write down the lower bound of the height.

..... cm
(1)

The weight of each cup is 120 g, correct to the nearest 10 g

(c) Write down the upper bound of the weight.

..... g
(1)

Working Unit 24

4MA1-2H Q22

UNIT 1
4 marks

- 22 The diagram shows a square inside rectangle $ABCD$

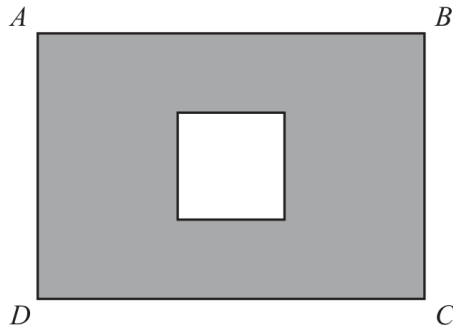


Diagram **NOT**
accurately drawn

The total area of the region shown shaded in the diagram is $X \text{ cm}^2$

$AB = 11.5 \text{ cm}$ correct to the nearest 0.5 cm

$BC = 9.2 \text{ cm}$ correct to 2 significant figures

side of square = 4.1 cm correct to 2 significant figures

By considering bounds, work out the value of X to a suitable degree of accuracy.
Show your working clearly.

$X = \dots\dots\dots$

(Total for Question 22 is 4 marks)

Question 26

4MA1-2H Q19

MODULAR UNIT 1

3 marks

19 $G = \frac{c}{2f - 3h}$

$c = 8$ correct to the nearest whole number

$f = 6.62$ correct to 2 decimal places

$h = 1.2$ correct to 1 decimal place

Work out the lower bound for the value of G

Give your answer correct to 3 decimal places.

Show your working clearly.

Question 29

4MA1-2H Q17 whole

MODULAR UNIT 1

3 marks

17 $P = a(c + y)$

 $a = 8.3$ correct to 2 significant figures $c = 2$ correct to 1 significant figure $y = 15$ correct to the nearest 5Work out the upper bound for the value of P

Show your working clearly.

.....

Question 29

4MA1-2H Q25

MODULAR UNIT 1

4 marks

- 25 A solid sphere has a radius of 2.8 centimetres, correct to 1 decimal place.
The sphere has a mass of $M\pi$ grams, where $M = 260$ correct to 2 significant figures.

Work out the upper bound for the density of the sphere.

Give your answer in g/cm^3 correct to 2 decimal places.

Show your working clearly.

..... g/cm^3

Question 16

4MA1-1H Q21 M01

MODULAR UNIT 1

3 marks

21 $T = \frac{x^2 + y^2}{w}$

$x = 28.4$ correct to 1 decimal place.

$y = 17$ correct to 2 significant figures.

$w = 90$ correct to the nearest 5

Calculate the upper bound for the value of T

Give your answer correct to 3 significant figures.

Show your working clearly.

.....

Question 18

4MA1-2H Q2 M01

MODULAR UNIT 1

1 marks

2 The length of a table is measured as 1.4 metres correct to one decimal place.

(a) Write down the upper bound of the length of the table.

..... metres

(1)

Question 19

4MA1-2H Q2 M02

MODULAR UNIT 1

1 marks

2 The length of a table is measured as 1.4 metres correct to one decimal place.

(b) Write down the lower bound of the length of the table.

..... metres

(1)

Question 25

4MA1-2H Q18 Q18

MODULAR UNIT 1

3 marks

18 $X = \frac{2a - b}{f}$

$a = 7.5$ correct to 1 decimal place.

$b = 3.42$ correct to 2 decimal places.

$f = 2$ correct to the nearest whole number.

Work out the upper bound of the value of X

Show your working clearly.

Question 15

4MA1-1H Q18

MODULAR UNIT 1

3 marks

18 $A = w - \frac{x^2}{y}$

$w = 3.45$ correct to 2 decimal places.

$x = 1.9$ correct to 1 decimal place.

$y = 5$ correct to the nearest whole number.

Work out the lower bound of the value of A

Show your working clearly.

Question 21

4MA1-2H Q4 Q4(a)

MODULAR UNIT 1

1 marks

- 4 Each side of a regular octagon has a length of 18 mm, correct to the nearest 0.5 mm

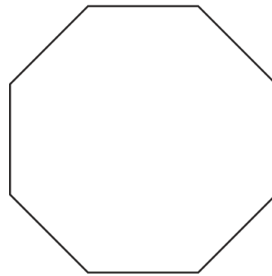


Diagram **NOT**
accurately drawn

- (a) Write down the lower bound of the length of each side of the octagon.

..... mm

(1)

Question 22

4MA1-2H Q4 Q4(b)

MODULAR UNIT 1

1 marks

(b) Write down the upper bound of the length of each side of the octagon.

..... mm

(1)

Give your answer correct to 2 decimal places.

$$a = \frac{v - u}{t}$$

$t = 1.84$ correct to 2 decimal places

Show your working clearly.

Calculate the upper bound for the value of P .
Show your working clearly.

A triangle with vertices X , Y , and Z . The interior angle at vertex X is 130° . The interior angle at vertex Y is 25° . The side XZ is labeled 16 cm .

The length XZ and the angles YXZ and XYZ are each given correct to 2 significant figures.

Calculate the upper bound for the length YZ .

Give your answer correct to one decimal place.

Show your working clearly.

..... cm

..... kg/m³

(Total for Question 17 is 4 marks)

$$a = \frac{v - u}{t}$$

$t = 8.5$ correct to 2 significant figures

By considering bounds, work out the value of a to a suitable degree of accuracy. Show your working clearly and give a reason for your answer.

(a) Write down the lower bound of the weight of the cake.

$$\dots\dots\dots \text{kg}$$

(1)

$$\dots\dots\dots \text{kg}$$

(1)

Penny's answer is not sensible.

(c) By rounding each number to one significant figure, work out a suitable estimate to show that her answer is not sensible.
Show your working clearly.

- $$\frac{2.11^2 \times 58.9}{\sqrt{8.859}}$$

(Total for Question 2 is 2 marks)

7 The weight of a bag of apples is 475 g correct to the nearest g

(a) Write down the lower bound of the weight.

..... g
(1)

The height of a box is 120 cm correct to the nearest 10 cm

(b) Write down the upper bound for the height.

..... cm
(1)

(Total for Question 7 is 2 marks)

Chapter OL-T09: Rounding, Estimation & Bounds

$n = 10.3$ correct to 1 decimal place
 $p = 7.24$ correct to 2 decimal places
 $q = 4.39$ correct to 2 decimal places

By considering bounds, work out the value of D to a suitable degree of accuracy. Show your working clearly.